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Unveiling the single-solvent vacuum-flash process for high-efficiency perovskite solar cells

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High-crystallinity perovskite films with low trap density are a prerequisite for realizing high-performance perovskite solar cells (PSCs). Here, we present a straightforward single-solvent vacuum-flash (SSVF) method for fabricating high-quality perovskite films using a simplified, single-solvent precursor system (e.g., *N,N*-dimethylformamide (DMF), dimethylacetamide (DMAC), and γ -valerolactone (GVL)). By creating a low temperature (LT) and vacuum environment for the precursor wet film, the SSVF method enables precise regulation of solvent evaporation dynamics, resulting in the formation of highly compact perovskite films with low trap densities. As a result, the PSCs fabricated via this method achieve a champion power conversion efficiency (PCE) of 26.93%, along with a certified PCE of 26.78% (certified steady-state efficiency of 26.39%). This performance ranks among the highest efficiencies ever reported for any vacuum-flash (VF)-based technique. Furthermore, owing to the enhanced phase stability of the perovskite film prepared via the SSVF method, the resulting PSC exhibits excellent operational durability, maintaining over 94% of its initial efficiency after 1000 hours of continuous operation.

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Broader context

To address global climate change and enable a sustainable energy transition, the development of next-generation photovoltaic technologies that are efficient, low-cost, and scalable is paramount. Perovskite solar cells (PSCs) are a leading contender, yet their commercialization has been persistently hindered by complex fabrication processes, particularly the reliance on toxic anti-solvents and hard-to-control solvent mixtures like DMSO/DMF. These traditional methods induce film defects and harmful residues, severely limiting large-area uniformity. Our work introduces a single-solvent vacuum-flash (SSVF) strategy that fundamentally overcomes these bottlenecks by precisely controlling solvent dynamics under a low temperature and vacuum condition. It offers a feasible approach for modulating crystallization pathways through physical means, which can provide valuable insights for the fabrication of other solution-processed functional films. Concurrently, by simplifying the process and addressing challenges associated with large-area fabrication, we hope to present a viable reference for advancing the scalability of perovskite solar cells. Furthermore, this method attempts to eliminate toxic solvents and has been successfully applied to green solvents such as γ -valerolactone (GVL), representing a promising direction for developing more environmentally benign and cost-effective manufacturing protocols.

Introduction

Lead halide perovskite materials have recently garnered significant attention within the photovoltaic community due to their outstanding semiconductor characteristics. These properties

include a high light absorption coefficient, high charge-carrier mobility, long carrier lifetimes, and a widely tunable bandgap.^{1–3} At present, the certified power conversion efficiency (PCE) of single-junction perovskite solar cells (PSCs) has reached 27.3%, a figure comparable to that of their crystalline silicon counterparts.⁴

The optoelectronic characteristics of PSCs are intrinsically linked to the quality of the crystalline absorber layer.^{5,6} Current methods for constructing PSCs, like the two-step deposition and anti-solvent techniques, face significant hurdles for industrial use.^{7–10} The two-step method often results in the presence of residual lead halide, which in turn impedes the device performance.^{11–13} The anti-solvent method, despite its ability

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to generate high-efficiency solar cells, suffers from several critical drawbacks. It has an extremely narrow processing time frame, is exceedingly sensitive to environmental factors (e.g., temperature and humidity), and relies heavily on the use of toxic anti-solvents. These characteristics collectively make it challenging to scale up the production process and pose safety risks.^{14–16} A highly promising alternative is the vacuum-flash (VF) method, which employs a vacuum to swiftly and uniformly evaporate the solvent from perovskite precursor films, thereby fostering the growth of high-quality perovskite crystal thin films.^{17–23} Although the VF method offers a safer, more reproducible, and scalable option by eliminating the need for toxic anti-solvents, the intricate nature of the precursor's solvent system continues to pose a significant challenge.^{21,24}

To achieve precise control over the crystallization process, researchers often employ complex solvent mixtures comprising two or more high-boiling-point solvents, such as dimethyl sulfoxide (DMSO), γ -butyrolactone (GBL), *N,N*-dimethylformamide (DMF), *N*-methylpyrrolidone (NMP), and 2-methoxyethanol (2-ME). During rapid solvent removal, the interplay between the differential evaporation rates of these solvents, their coordination affinity with lead ions, and the decomposition pathways of intermediate phases makes it exceedingly difficult to predict the precise impact of adjusting solvent ratios or annealing profiles on the resulting film morphology and optoelectronic performance.^{25–27} Although the DMSO and DMF co-solvent system is widely regarded as the “gold standard” for fabricating high-efficiency PSCs,^{8,14,28–30} its inherent drawbacks pose a significant obstacle to the commercialization of this technology. The detrimental effects of DMSO, which acts as a “double-edged sword”, primarily stem from three key aspects. Firstly, its high boiling point (189 °C) and strong coordination ability lead to unavoidable solvent residues after annealing, particularly at the crucial buried interface.^{31–33} This residual solvent directly impairs the thermal stability and long-term operational lifetime of the device. Secondly, being a potent Lewis base, DMSO establishes excessively stable intermediate adducts with lead iodide (PbI₂).^{34,35} When subjected to rapid or inadequate annealing conditions, the phase transformation of these adducts fails to reach completion, leading to their entrapment within the film as defects.^{31,36} These defects act as non-radiative recombination centers, thereby limiting the open-circuit voltage (V_{oc}) and the overall performance of the device. Lastly, the pronounced hygroscopic nature of DMSO makes it easy for ambient moisture to penetrate. This moisture ingress triggers the degradation of the photoactive perovskite phase, transforming it into an inactive phase.³⁷ At the device level, these problems are evident as increased defect densities, shortened carrier lifetimes, and poor performance, particularly in large-area solar cells.³⁸

Herein, we report a single-solvent vacuum-flash (SSVF) method that enables the fabrication of high-performance PSCs. This method hinges on lowering the operating temperature to precisely govern the solvent evaporation dynamics. By doing so, it offers fine-grained control over the crystallization pathway of the perovskite. This strategy has successfully resulted in the formation of highly dense perovskite films. More importantly,

it provides an excellent platform for elucidating the fundamental film-formation mechanism of this simplified system. Benefiting from the superior film quality, our devices achieved a champion PCE of 26.93% (certified PCE of 26.78% with a steady-state efficiency of 26.39%). Encouragingly, devices fabricated with a larger aperture area of 1.01 cm² retained a remarkable PCE of 25.81%. In addition to that, the encapsulated device retained over 94% of its initial performance after 1000 hours of stabilized power output (SPO) measurement in air (40 ± 5 °C, 35%–45%RH).

Results and discussion

Fig. 1a schematically illustrates the fabrication process for perovskite films *via* our single-solvent vacuum-flash (SSVF) method. First, a uniform wet perovskite film was prepared *via* spin-coating. Following this, the wet film was cooled on a coldplate for a defined period before excess solvent was rapidly removed using a VF evaporation system. Since the rate of solvent evaporation is significantly reduced at a low temperature (LT) (Fig. S1, SI), the use of a coldplate to cool the wet film enables precise control over solvent evaporation kinetics, which is discussed in detail below. Finally, the film underwent annealing on a hotplate to yield the final perovskite film. To highlight the broad applicability of our single-solvent flash method, we successfully prepared four distinct perovskite compositions, including FA_{0.95}CS_{0.05}PbI₃, FAPbI₃, FA_{0.8}MA_{0.15}CS_{0.05}Pb(I_{0.75}Br_{0.25})₃, and FA_{0.8}CS_{0.2}Pb(I_{0.6}Br_{0.4})₃. Scanning electron microscopy (SEM) images (Fig. 1b and c) reveal that all compositions yield highly compact and uniform films with a void-free buried interface. X-ray diffraction (XRD) analysis (Fig. S2, SI) confirms the successful formation of the photoactive black-phase perovskite for all four compositions fabricated using this process.

To elucidate the underlying mechanism of the SSVF process, we further investigated the influence of wet film temperature and vacuuming duration on the resulting film morphology. Fig. 2a presents a comparative analysis of digital photographs showing VF-processed precursor films for FA_{0.95}CS_{0.05}PbI₃ perovskite, where a progressive reduction in film transparency and a concurrent darkening of color are observed with increasing coldplate temperature. Films prepared at room temperature (RT) immediately turned brown and exhibited a non-uniform surface gloss. The observed reduction in film transparency at elevated temperatures stems from increased void formation due to accelerated DMF evaporation, while the concurrent color change (from transparent to dark brown) suggests a perovskite phase transition. Then, XRD measurement was performed on these five films to understand their precursor compositions, with the resulting patterns shown in Fig. 2b. The data indicate that for coldplate temperatures below 2 °C, the XRD patterns exclusively show diffraction peaks corresponding to the δ -phase. Notably, when the processing temperature exceeds 6 °C, a distinct α -phase diffraction peak emerges in the XRD pattern. The precursor film obtained by VF treatment at RT displays a significantly

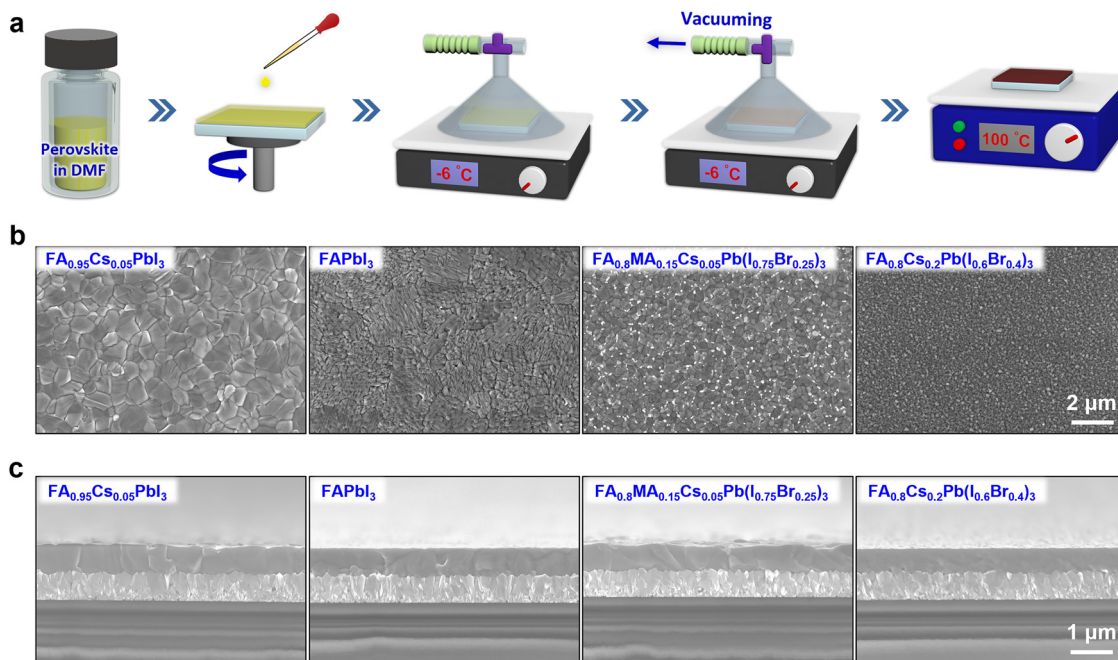


Fig. 1 (a) Schematic process of the SSVF method. (b) Top-view and (c) cross-sectional SEM images of the $\text{FA}_{0.95}\text{Cs}_{0.05}\text{PbI}_3$, FAPbI_3 , $\text{FA}_{0.8}\text{MA}_{0.15}\text{Cs}_{0.05}\text{Pb}(\text{I}_{0.75}\text{Br}_{0.25})_3$, and $\text{FA}_{0.8}\text{Cs}_{0.2}\text{Pb}(\text{I}_{0.6}\text{Br}_{0.4})_3$ perovskite films prepared based on the SSVF method.

higher signal intensity for the α -phase compared to that of the δ -phase. This suggests a propensity for the perovskite system to form the δ -phase under low-temperature vacuum conditions, while higher temperatures promote a transformation to the α -phase. Interestingly, no diffraction signals corresponding to any intermediate phase were detected in these five samples, which would seemingly indicate the complete removal of DMF from the precursor film after the 40-second VF treatment. To probe for residual DMF, precursor films prepared by the VF method at -6°C and RT were analyzed by Fourier transform infrared (FTIR) spectroscopy (Fig. 2c). The film prepared at RT exhibited no distinct $\text{C}=\text{O}$ absorption peak in the FTIR spectrum, confirming effective removal of DMF solvent. The film prepared at -6°C , however, features a prominent peak at 1649 cm^{-1} corresponding to the $\text{C}=\text{O}$ bond, clearly evidencing the presence of residual DMF. Accordingly, we monitored the film prepared *via* LTVF at -6°C by XRD after resting at RT for varying durations (Fig. 2d). The resulting patterns show diffraction signals at 2θ of 6.5° , 8.0° , and 9.5° , which appear and then dissipate. This phenomenon persisted even when the vacuum treatment duration was extended to 180 s (Fig. S3, SI). Through a systematic investigation of the intermediate phase, we definitively assign the new XRD diffraction signals to the formation of the $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMF}$ intermediate phase (Fig. S4, SI). This observation further confirms the presence of a minor amount of residual DMF solvent within the precursor film. Subsequently, upon annealing the precursor film at 60°C , a clear transition from the δ -phase to the α -phase was observed (Fig. S5, SI).

Top-view SEM images (Fig. 2e) reveal that perovskite films prepared under various coldplate temperatures exhibit a dense

morphology. However, cross-sectional SEM images (Fig. 2f) clearly indicate the formation of voids at the bottom of the perovskite film when the coldplate temperature exceeds 2°C . Furthermore, the number of these voids evidently increases with rising temperature. For comparison, the film prepared *via* the room-temperature VF method likewise maintains excellent surface coverage; however, a more pronounced formation of interfacial voids is evident at its bottom. Finally, the RT vacuum condition results in the perovskite films with significantly reduced crystallinity compared to those prepared by SSVF treatment (Fig. S6, SI). Furthermore, the impact of vacuum duration on perovskite film morphology was examined, revealing a gradual reduction in both pinholes and interfacial voids with longer treatment times (Fig. S7, SI). This is attributed to the shorter VF durations, which allow for the precipitation of only a thin precursor layer on the wet film surface while leaving a significant amount of DMF retained within the film bulk. Subsequent rapid heating to 100°C triggers secondary dissolution of the surface precursor layer by residual DMF, resulting in the formation of distinct voids within the perovskite film.³⁹ In contrast, the duration of VF treatment at RT has a negligible impact on film morphology (Fig. S8, SI), likely because DMF evaporates so rapidly under this condition that even a brief vacuum step is sufficient to remove most of the solvent, rendering any further extension of the vacuum treatment duration inconsequential.

Based on these results, we can infer a film formation mechanism for the VF treatment (Fig. 3a). First, under low-temperature and vacuum conditions, a capping layer, primarily consisting of the δ -phase, rapidly precipitates on the wet film surface. Due to the slower solvent volatilization rate at LTs,

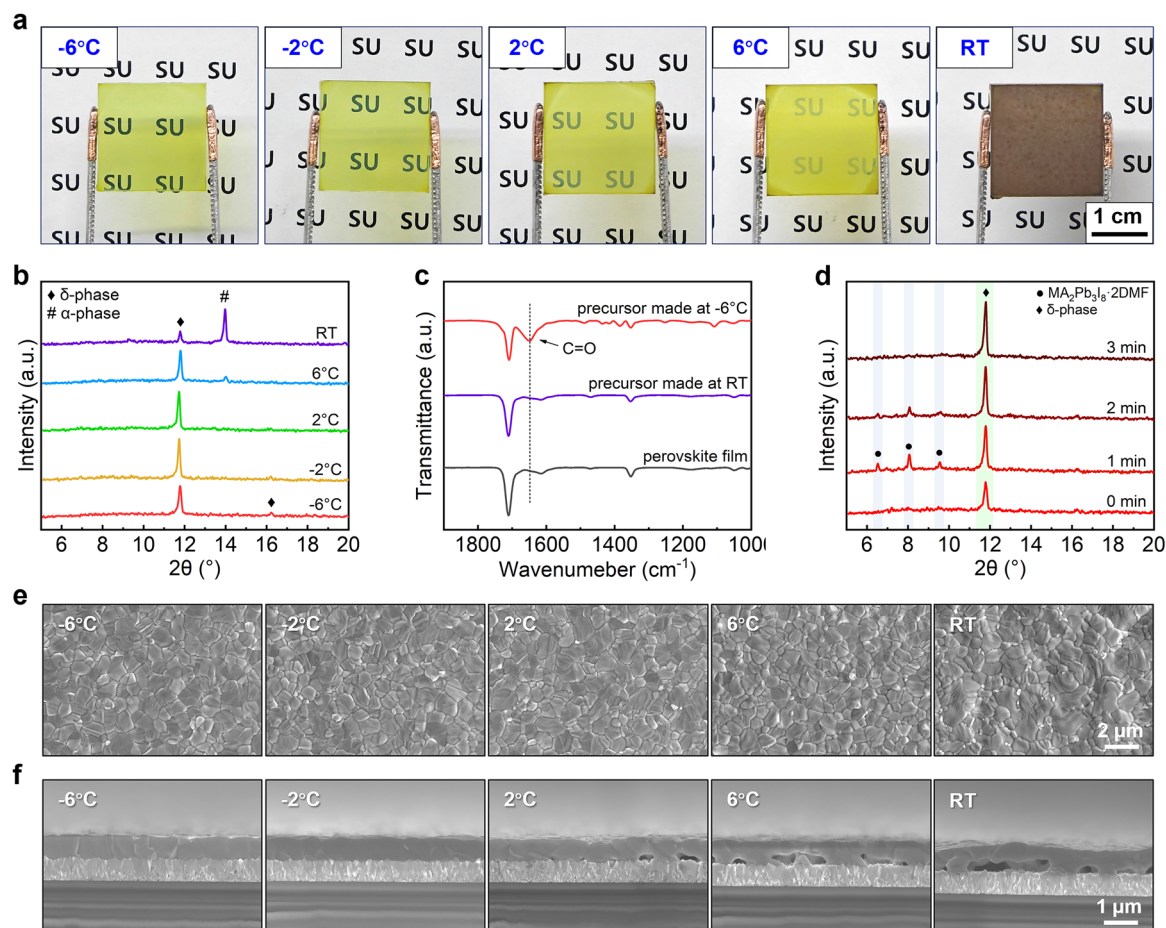


Fig. 2 (a) Digital photographs and (b) XRD patterns of the SSVF-processed perovskite precursor films at different coldplate temperatures (temperature error: ± 1 °C). (c) FTIR spectra of the different films prepared with the DMF-only solvent. (d) Evolving XRD pattern of the precursor film prepared by an SSVF treatment at -6 °C. (e) Top-view and (f) cross-sectional SEM images of the perovskite films prepared by SSVF treatment at different coldplate temperatures.

DMF within the wet film gradually escapes. As the thickness of the capping layer increases, the DMF escape rate continues to diminish. Consequently, after 40 s of LT-VF treatment, a small amount of residual solvent remains trapped within the precursor film. However, during the subsequent annealing step, the volatilization of this residual DMF is sufficiently gentle that it does not generate voids at the buried interface. XRD patterns (Fig. 2d and Fig. S3, SI) reveal that if the as-prepared precursor film is left at RT without annealing, residual DMF reacts with the surface capping layer to transiently form an intermediate phase. This phase subsequently decomposes, releasing the entrapped DMF (Fig. S9, SI). However, this RT aging process, which lasts for several minutes, has a negligible impact on the final perovskite film morphology (Fig. S10, SI). Based on these findings, we propose an empirical model to describe the solvent evaporation flux (J) during the vacuum-flash process as a function of the wet film temperature (T), vacuum pressure (P_{vac}), and treatment time (t) (Note S1, SI):

$$J(t, T, P_{\text{vac}}) = \sqrt{\frac{D(T) \cdot \Delta C(T, P_{\text{vac}})}{2k}} \cdot \frac{1}{\sqrt{t}} \quad (1)$$

where, D is the diffusion coefficient of the solvent within the capping layer, which is a key parameter that is strongly dependent on temperature; ΔC is the solvent concentration difference across the capping layer, a term influenced by both temperature and vacuum pressure; k is a proportionality constant related to the concentration and density of the solute in the initial solution. According to (eqn (1)), the solvent evaporation flux from a wet film at RT is significantly higher than at LTs (e.g., -6 °C). This excessively rapid solvent escape leads to the formation of voids (Fig. 3b). Conversely, if the solvent evaporation flux is too low (e.g., due to an insufficient vacuuming time or too low temperature), only a thin capping layer will precipitate on the wet film's surface, trapping a significant amount of solvent within the film bulk. If this film is then subjected to rapid high-temperature annealing, the entrapped residual solvent will diffuse upwards and penetrate the capping layer. The subsequent rapid escape of this solvent can cause secondary dissolution of the surface layer, leading to the formation of pinholes in the final film and compromising its coverage (Fig. 3c).

The broad applicability of our model is further underscored by its successful extension to other solvents with melting and

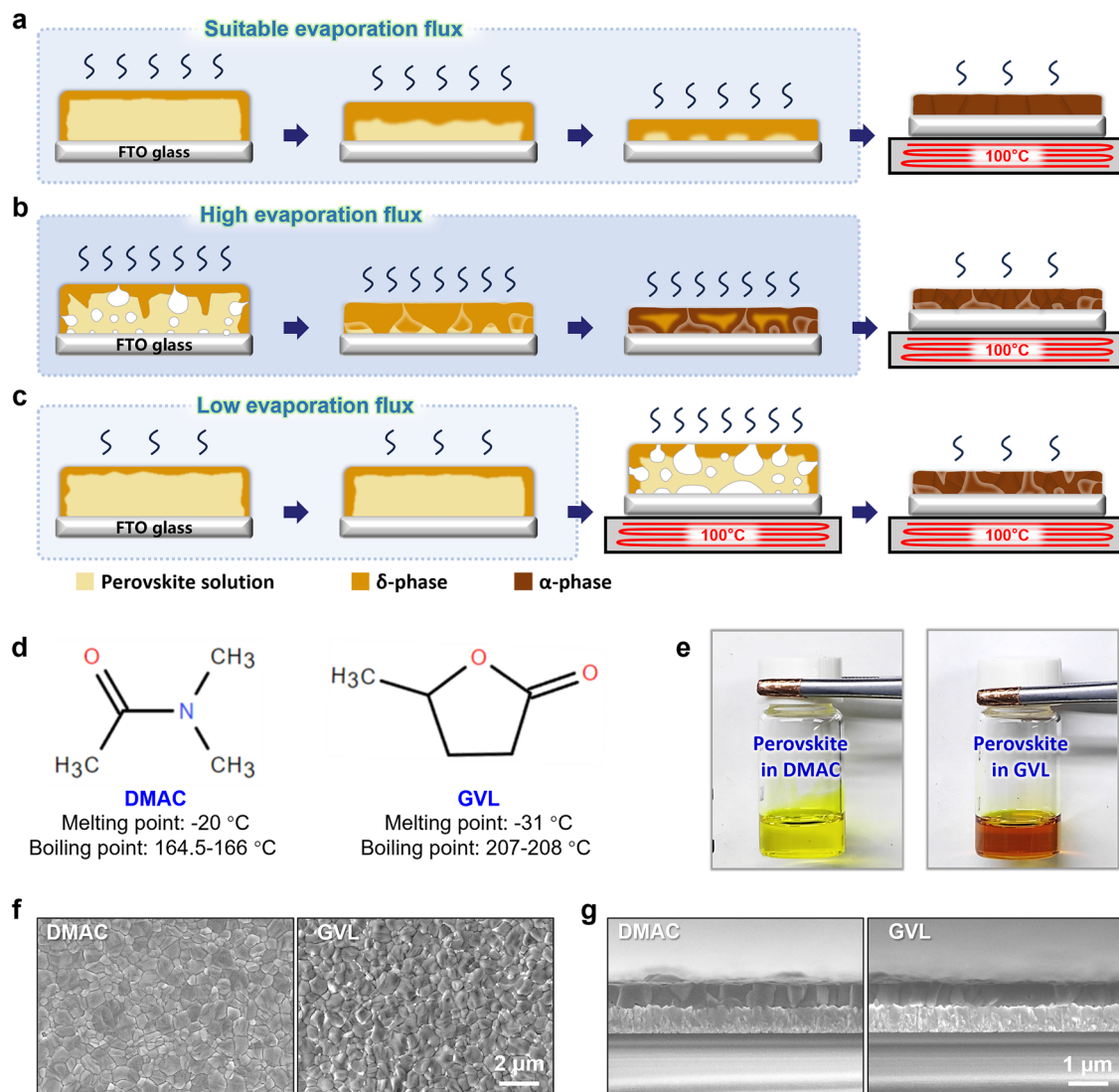


Fig. 3 Schematic evolution of the perovskite precursor film at (a) suitable, (b) high, and (c) low evaporation fluxes during the SSVF process. (d) Chemical formula of DMAC and GVL. (e) Digital photographs of the perovskite solutions prepared with DMAC- and GVL-only solvents, respectively. (f) Top-view and (g) cross-sectional SEM images of the perovskite films prepared by SSVF treatment from DMAC- and GVL-only solvents, respectively.

boiling points similar to DMF, such as dimethylacetamide (DMAC) and γ -valerolactone (GVL) (Fig. 3d and e). By optimizing the interplay between coldplate temperature and vacuum pressure, we produced uniform, highly transparent precursor films (Fig. S11, SI) that subsequently converted into high-quality, void- and pinhole-free perovskite layers. The optimized conditions were determined to be $-4\text{ }^{\circ}\text{C}$ and 10 Pa for DMAC, and $-6\text{ }^{\circ}\text{C}$ and 100 Pa for GVL, respectively (Fig. 3f and g). Furthermore, this framework explains the “whitening” phenomenon observed at the buried interface of perovskite films fabricated *via* the VF method using a DMSO–DMF mixed-solvent system. This whitening is indicative of void formation in this region, which is exacerbated at LTs (Fig. S12, SI). Indeed, FTIR spectroscopy confirms the presence of a pronounced DMSO signal in the VF-processed precursor films under both RT- and LT-VF conditions (Fig. S13, SI). Concurrently, X-ray diffraction (XRD) analysis of the precursor film (Fig. S14, SI)

reveals the preferential formation of a highly stable $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMSO}$ intermediate phase,^{32,34,39} which exhibits greater stability than its corresponding $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMF}$ counterpart (Fig. 2d). The remarkable stability of this DMSO-adduct is further evidenced by the persistence of its characteristic diffraction peaks even after the film was held at RT for 20 min. This solvent retention is attributed to the strong coordinating capacity of DMSO with PbI_2 , resulting in trapped solvent within the precursor film.³⁹ The subsequent volatilization of this entrapped DMSO during thermal annealing is ultimately responsible for the creation of interfacial voids. These findings also explain the inherent difficulty in achieving complete solvent removal under conventional processing conditions.

Building on these optimized conditions, we performed a systematic comparison of the morphology, crystallinity, and optical and electrical properties of perovskite films derived from DMSO–DMF, DMF, DMAC, and GVL solvents. The film

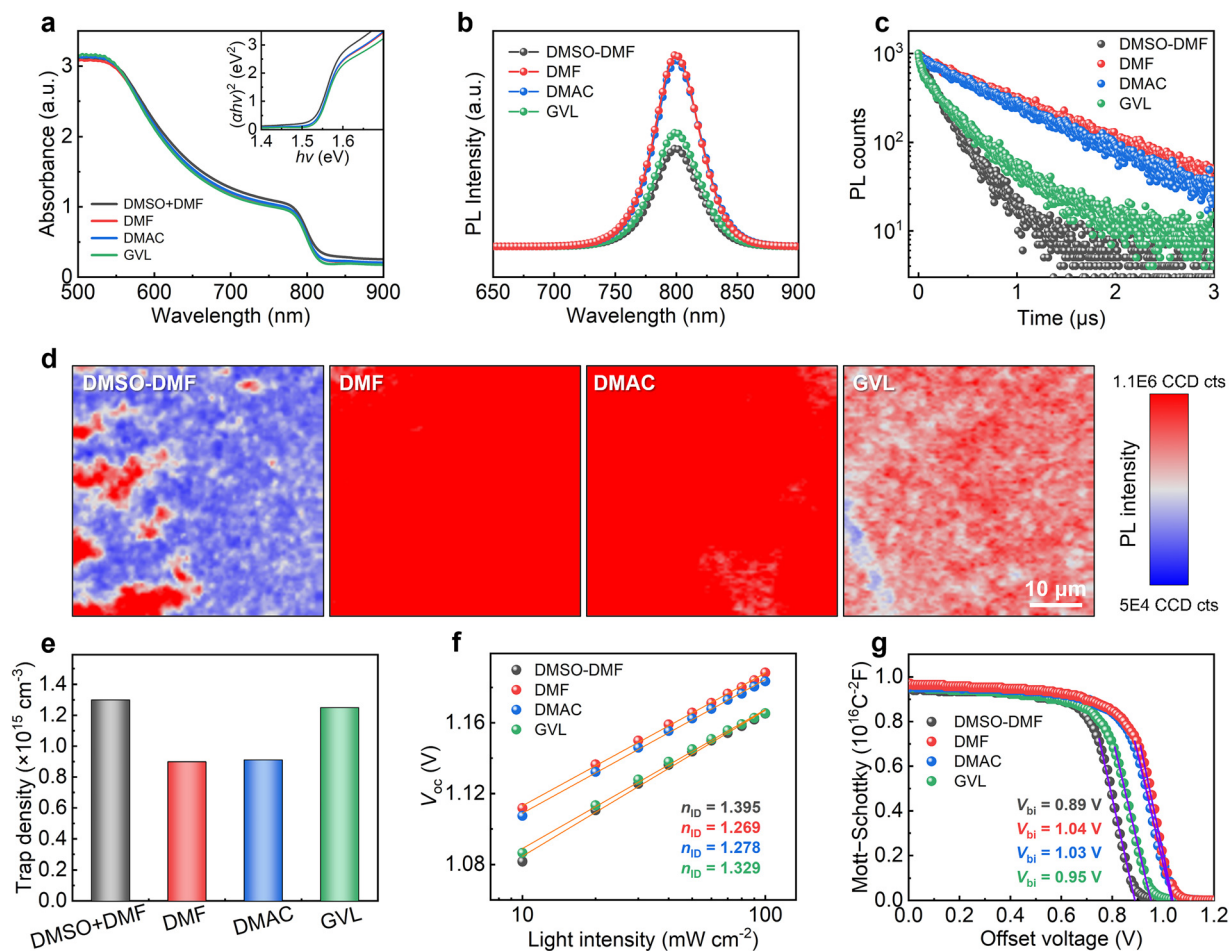


Fig. 4 (a) UV-visible absorption, (b) PL spectra, (c) TRPL spectra, and (d) PL mapping of the perovskite films prepared with DMSO–DMF, DMF, DMAC, and GVL solvents, respectively. (e) Trap densities calculated from the SCLC test for the perovskite films prepared with DMSO–DMF, DMF, DMAC, and GVL solvents, respectively. (f) Photovoltage as a function of the light intensity and (g) Mott–Schottky plots for the photovoltaic devices fabricated based on the perovskite films prepared with DMSO–DMF, DMF, DMAC, and GVL, respectively.

derived from the DMSO–DMF system shows inferior crystallinity, as evidenced by its significantly weaker XRD peak intensities compared to the other three single-solvent counterparts (Fig. S15, SI). Morphologically, although SEM imaging reveals a dense surface, some voids are observed at the film-substrate interface of the DMSO–DMF sample (Fig. S16, SI). The observed void formation is likely attributed to the entrapment of excess residual solvent from the VF step, which subsequently evaporates during thermal annealing, creating localized porosity in the film. Fig. 4a displays the ultraviolet-visible (UV-vis) absorption spectra of the perovskite films prepared with the four distinct solvent protocols. The corresponding Tauc plots, derived from the absorption spectra (inset of Fig. 4a), reveal that all the films exhibit an optical bandgap of approximately 1.53 eV. Photoluminescence (PL) spectra (Fig. 4b) and PL mapping (Fig. 4d) show that perovskite films prepared with single-solvent systems not only possess stronger PL intensity but also exhibit a more uniform PL distribution. Concurrently, time-resolved photoluminescence (TRPL) spectra (Fig. 4c) indicate that perovskite films prepared with DMSO–DMF solvent

have the fastest PL decay rate. To quantitatively analyze the carrier dynamics, the TRPL curves were fitted using a bi-exponential decay function.^{40,41} The average carrier lifetimes (τ_{avg}) were calculated from the fitting parameters and are summarized in Table S1 in SI. Notably, the perovskite film prepared using the DMSO–DMF solvent demonstrates a substantially shorter average carrier lifetime (229.0 ns) compared to the films made with the DMF (945.9 ns), DMAC (831.9 ns), and GVL (344.0 ns) systems. These findings collectively suggest that the single-solvent system is conducive to reducing non-radiative recombination within the perovskite films. Furthermore, we fabricated electron-only devices based on perovskite films prepared with different solvent systems and conducted space-charge-limited current (SCLC) tests (Fig. S17, SI). The calculated trap densities (Fig. 4e) clearly demonstrate that perovskite films prepared with single solvents possess lower trap densities.

To elucidate the impact of the improved film quality on device performance, we fabricated PSCs based on the films prepared with different solvent systems. To probe the dominant

recombination mechanisms within the finished devices, the dependence of the V_{oc} on light intensity (P_{light}) was systematically investigated (Fig. 4f). The diode ideality factor (n_{ID}) was extracted by fitting the semi-logarithmic plot of V_{oc} versus P_{light} according to the relationship: $V_{oc} = (n_{ID}k_B T/q) \ln P_{light} + \text{constant}$, where k_B is the Boltzmann constant, T is temperature, and q represents the elementary charge.^{42,43} An n_{ID} value closer to 1 signifies the dominance of bimolecular recombination, whereas a value closer to 2 indicates that trap-assisted Shockley–Read–Hall (SRH) recombination is the primary carrier loss pathway. Notably, the devices fabricated with DMF-, DMAC-, and GVL-only solvents exhibited superior diode quality, with n_{ID} values calculated to be 1.269, 1.278, and 1.329, respectively. This is substantially lower than the value (1.395) obtained for devices made from the DMSO–DMF solvent system, suggesting a significant suppression of trap-assisted non-radiative recombination. This finding demonstrates excellent agreement with the prolonged carrier lifetimes observed *via* TRPL, confirming that the optimized crystallization process effectively reduced both bulk and interfacial defects in the perovskite film. Consequently, a superior V_{oc} and fill factor (FF) are anticipated for these devices. To further investigate the origin of the potential V_{oc} enhancement, we performed Mott–Schottky measurements to determine the built-in potential (V_{bi}) across the device junctions. As shown by the C^{-2} versus voltage plots (Fig. 4g), the V_{bi} is extrapolated from the x -intercept.^{41,44} The results clearly indicate that devices fabricated with single solvents achieved substantially larger V_{bi} values. This is a marked improvement over the devices prepared with DMSO–DMF solvent. A larger V_{bi} generates a stronger electric field within the depletion region, which facilitates more efficient charge carrier separation and collection while simultaneously increasing the

maximum achievable photovoltage. Therefore, with both reduced non-radiative recombination and an enhanced built-in potential, devices constructed with the single solvent are fundamentally poised to deliver superior photovoltaic performance.

Leveraging the superior film quality achieved with the SSVF method, we assembled and characterized the photovoltaic performance of p–i–n PSCs. Fig. 5a provides a schematic illustration of the device structure: FTO/SAM/perovskite/passivation layer/ C_{60} /SnO₂/Ag. The J – V statistical results for small-area devices (0.107 cm²) (Fig. 5b) demonstrate that devices prepared with single solvents exhibit higher V_{oc} s and FFs. Notably, the optimal device fabricated using the DMF-only solvent achieves a PCE of 26.93% (Fig. 5c), with a certified PCE of 26.78% (certified steady-state efficiency $\sim$ 26.39%) from the National Photovoltaic Industry Metrology and Testing Center (NPVM) (Fig. S18, SI). This represents one of the highest efficiencies reported to date for PSCs fabricated using a VF treatment process. Furthermore, using our SSVF method, we achieve outstanding power conversion efficiencies (PCEs) of 26.88% and 26.27% for PSCs fabricated with DMAC-only and GVL-only solvents, respectively (Fig. S19, SI). We highlight the result with GVL, as it is a recognized non-toxic, green solvent. Our ability to attain a PCE of 26.27% with this environmentally benign solvent demonstrates the excellent compatibility of the VF process for various perovskite compositions and solvent systems. Fig. 5d shows the external quantum efficiency (EQE) spectrum of the champion device made from a DMF-only solvent system. It can be observed that this PSC exhibits EQE values exceeding 90% across the visible light spectrum, with an integrated current density of 25.69 mA cm^{−2}, which matches the J_{sc} obtained from J – V measurements well. Remarkably, upon scaling up the cell area to 1.01 cm², a PCE of 25.81%

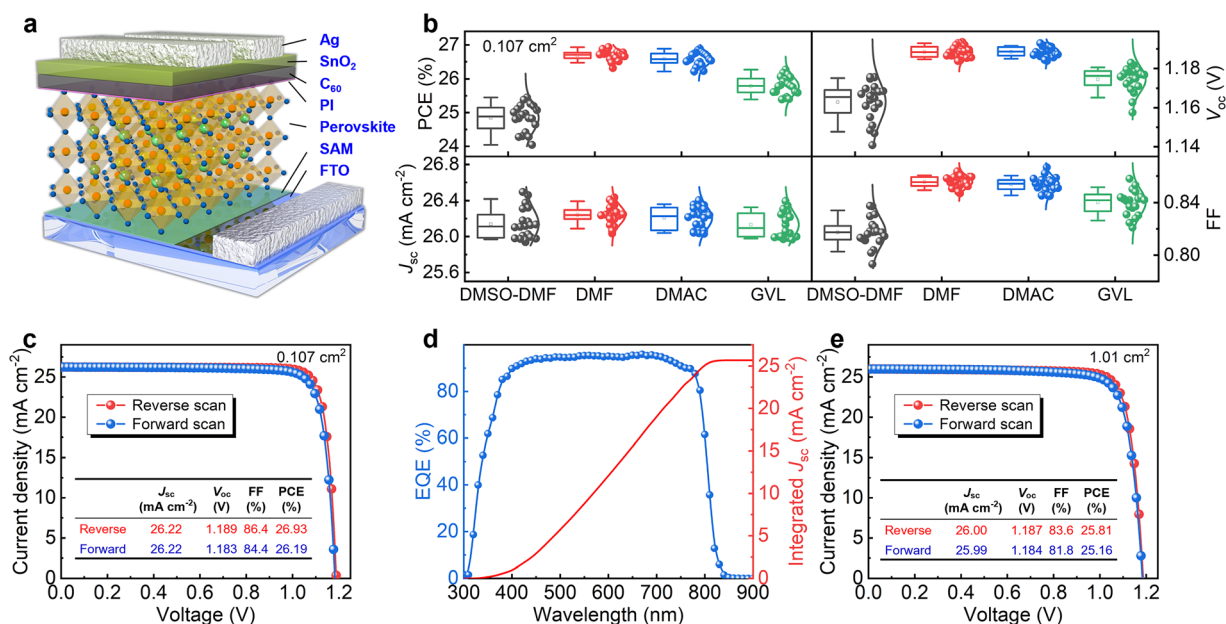


Fig. 5 (a) Schematic architecture of the p–i–n PSCs. (b) Statistical PCEs of the 0.107-cm² PSCs fabricated with different solvent systems. (c) J – V curves and (d) EQE spectra of the 0.107-cm² champion device. (e) J – V curves of the 1.01-cm² champion device.

(average PCE of 25.53%) is achieved (Fig. 5e and Fig. S20 in SI). This is currently the highest reported efficiency for 1-cm² PSCs in the literature (Fig. S21 and Table S2 in SI). Furthermore, the minimal efficiency difference between small-area and large-area PSCs fabricated using the single-solvent VF method in a DMF-only system indicates the ability of this technique to mitigate performance losses associated with increased device area.

Finally, we systematically investigate the phase stability of the perovskite films and the operational stability of corresponding devices prepared from the DMSO–DMF and DMF-only solvent systems. To assess phase stability, we subject the films to thermal aging at 85 °C in a N₂ atmosphere and characterize them at various time points. The SEM imaging (Fig. 6a) and

XRD analysis (Fig. 6c and Fig. S22a in SI) reveal that the DMSO–DMF-based film exhibits PbI₂ precipitation after only 48 h of heating, with the amount increasing over time. In stark contrast, we observe only trace amounts of PbI₂ in the DMF-only film even after 432 h of continuous heating (Fig. 6b, d, and Fig. S22b in SI). Furthermore, the PL emission peak of the DMSO–DMF film exhibits a significant blueshift and broadening during aging, whereas the DMF-only film maintains a highly stable PL profile (Fig. 6e, f and Fig. S23 in SI). We attribute this instability in the DMSO–DMF film to two primary factors. First, the volatilization of organic components leads to the precipitation of PbI₂, which can reduce the effective grain size and increase grain boundaries, thereby inducing a quantum confinement effect that widens the bandgap. Second,

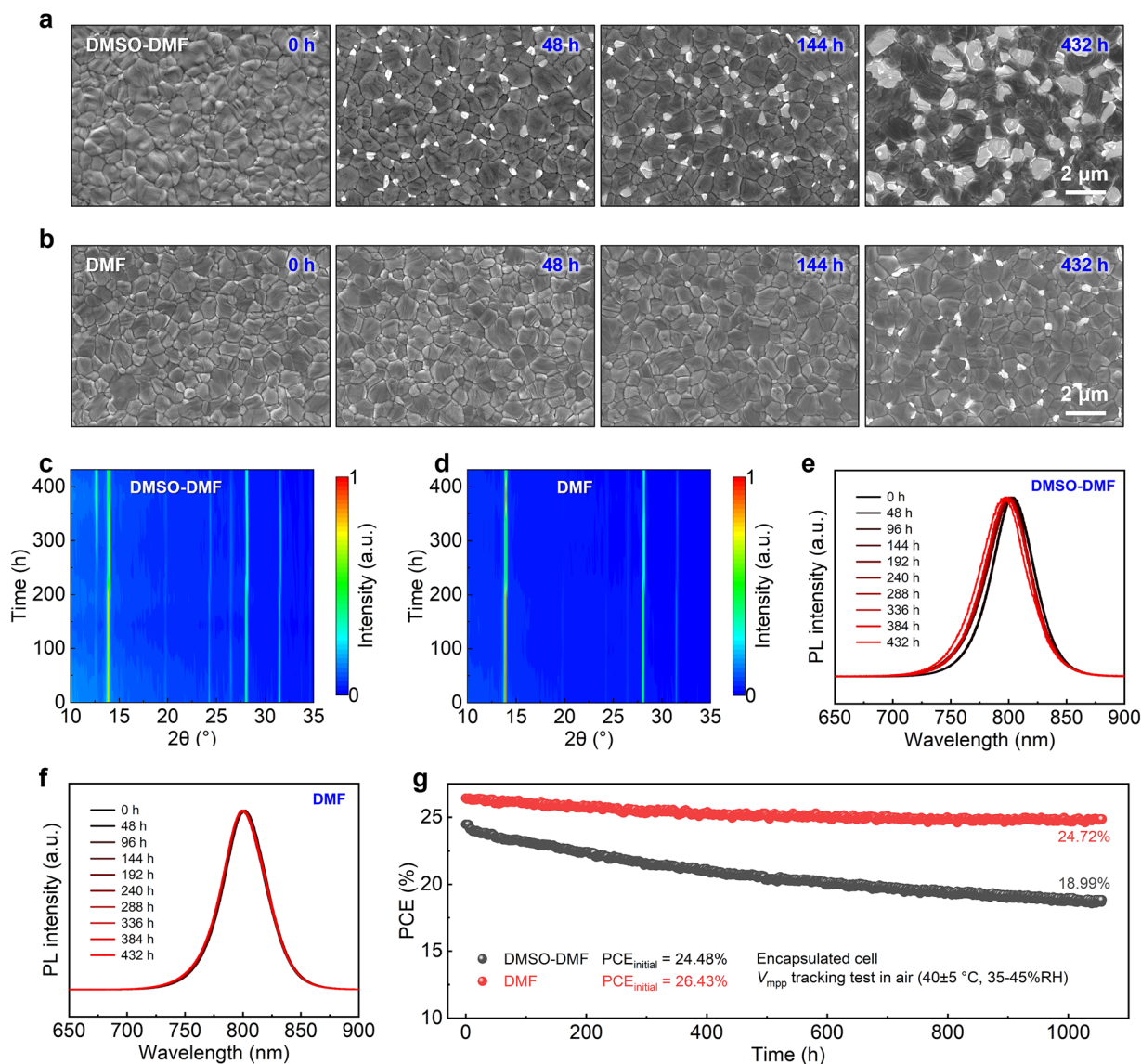


Fig. 6 SEM images of the perovskite films prepared with (a) DMSO–DMF and (b) DMF solvent systems at various 85 °C heating periods. XRD patterns of the perovskite films prepared with (c) DMSO–DMF and (d) DMF solvent systems as a function of heating times. PL spectra of the perovskite films prepared with (e) DMSO–DMF and (f) DMF solvent systems at various heating periods. (g) MPP tracking of the PSCs fabricated with DMSO–DMF and DMF-only solvent systems, respectively.

the loss of FA⁺ cations increases the relative Cs⁺/FA⁺ ratio in the film, causing a contraction of the perovskite lattice that also results in a wider bandgap. The superior phase stability of the DMF-only film, which we attribute to its enhanced crystallinity, effectively suppresses these degradation pathways. In addition to film stability, we compare the operational stability of encapsulated PSCs. From the maximum power point (MPP) tracking data (Fig. 6g), it can be observed that the DMF-only device maintains over 94% of its initial efficiency after 1000 hours of continuous operation in air, whereas the device fabricated with the DMSO–DMF co-solvent exhibits more significant degradation, retaining only 77.6% of its initial performance.

Conclusions

In summary, we have developed a facile SSVF strategy for fabricating high-quality perovskite films. By lowering the wet-film temperature, the SSVF method enables precise control over the solvent volatilization kinetics. This controlled process prevents the formation of interfacial voids commonly observed in RT processes and yields exceptionally compact, crystalline perovskite films with low trap densities. Benefiting from this superior film quality, the SSVF-fabricated devices demonstrated a champion PCE of 26.93% (certified at 26.78%) for small-area cells (0.107 cm²). Crucially, the process shows exceptional scalability, with large-area (1.01 cm²) devices achieving a very high PCE of 25.81%. This work offers critical insights into precise crystallization control and provides a viable pathway for the scalable and robust manufacturing of high-performance PSCs.

Author contributions

H. Z. and J. P. conceived the idea. H. Z., Y. P., and Y. Z. performed the primary experiments and data analysis. Y. W. and Z. F. assisted with digital photography. Y. G., H. H., and C. G. contributed to the XRD and SEM analyses. X. L., X. W., and Y. L. collected the FTIR and UV-vis absorption spectra. C. H., Y. Z., and J. S. conducted the PL and TRPL measurements. H. Z. wrote the initial manuscript, with Y. P., Y. J., and J. P. contributing to the review and editing. H. Z., Y. J., X.-H. Z., and J. P. supervised the project. All authors discussed the results and commented on the manuscript.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information: Experimental Section, Note S1, Fig. S1–S23, and Tables S1–S2. See DOI: <https://doi.org/10.1039/d5ee05421d>.

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